

Elements of Numerical Integration

(Sayisal integral)

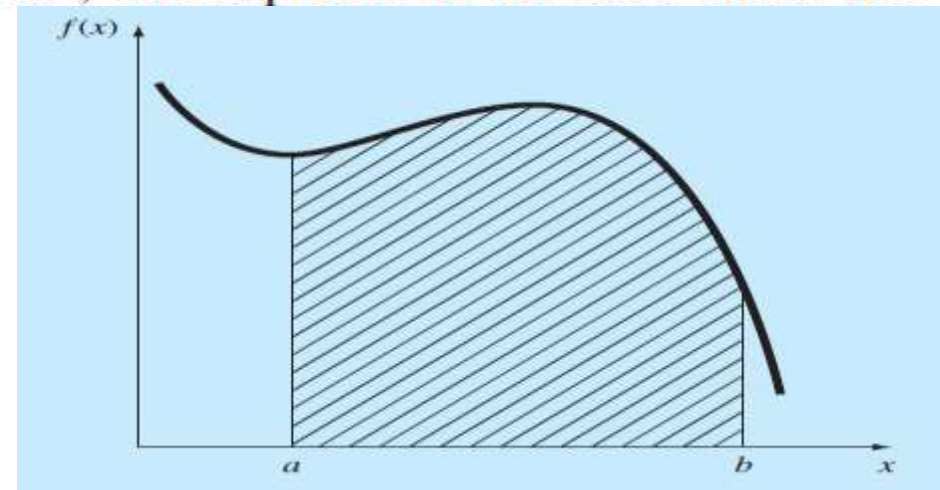
The inverse process to differentiation in calculus is integration. According to the dictionary definition, to *integrate* means “to bring together, as parts, into a whole; to unite; to indicate the total amount” Mathematically, integration is represented by

$$I = \int_a^b f(x) dx \quad (\text{PT6.2})$$

which stands for the integral of the function $f(x)$ with respect to the independent variable x , evaluated between the limits $x = a$ to $x = b$. The function $f(x)$ in Eq. (PT6.2) is referred to as the *integrand*.

As suggested by the dictionary definition, the “meaning” of Eq. (PT6.2) is the *total value*, or *summation*, of $f(x) dx$ over the range $x = a$ to b . In fact, the symbol $\int$ is actually a stylized capital S that is intended to signify the close connection between integration and summation.

Figure PT6.2 represents a graphical manifestation of the concept. For functions lying above the x axis, the integral expressed by Eq. (PT6.2) corresponds to the area under the curve of $f(x)$ between $x = a$ and b .¹



The need often arises for evaluating the definite integral of a function that has no explicit antiderivative or whose antiderivative is not easy to obtain. The basic method involved in approximating $\int_a^b f(x) dx$ is called **numerical quadrature**. It uses a sum $\sum_{i=0}^n a_i f(x_i)$ to approximate $\int_a^b f(x) dx$.

The methods of quadrature in this section are based on the interpolation polynomials given in Chapter 3. The basic idea is to select a set of distinct nodes $\{x_0, \dots, x_n\}$ from the interval $[a, b]$. Then integrate the Lagrange interpolating polynomial

$$\underbrace{P_n(x)}_{\uparrow} = \sum_{i=0}^n f(x_i) L_i(x) \quad \rightarrow L_0 f(x_0) + L_1 f(x_1) + \dots$$

and its truncation error term over $[a, b]$ to obtain

error formula: *hata formulü*

$$\begin{aligned} \boxed{\int_a^b f(x) dx} &= \int_a^b \sum_{i=0}^n f(x_i) L_i(x) dx + \int_a^b \prod_{i=0}^n (x - x_i) \frac{f^{(n+1)}(\xi(x))}{(n+1)!} dx \\ &= \sum_{i=0}^n \underbrace{a_i}_{\checkmark} f(x_i) + \frac{1}{(n+1)!} \int_a^b \prod_{i=0}^n (x - x_i) f^{(n+1)}(\xi(x)) dx, \end{aligned}$$

a < \xi < b

where $\xi(x)$ is in $[a, b]$ for each x and

$$a_i = \int_a^b L_i(x) dx, \quad \text{for each } i = 0, 1, \dots, n.$$

The quadrature formula is, therefore,

$$\int_a^b f(x) dx \approx \sum_{i=0}^n a_i f(x_i),$$

with error given by

$$E(f) = \frac{1}{(n+1)!} \int_a^b \prod_{i=0}^n (x - x_i) f^{(n+1)}(\xi(x)) dx.$$

Before discussing the general situation of quadrature formulas, let us consider formulas produced by using first and second Lagrange polynomials with equally-spaced nodes. This gives the **Trapezoidal rule** and **Simpson's rule**

The Trapezoidal Rule (Yamuk Kuralı)

To derive the Trapezoidal rule for approximating $\int_a^b f(x) dx$, let $x_0 = a$, $x_1 = b$, $h = b - a$ and use the linear Lagrange polynomial:

$$P_1(x) = \frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1).$$

Then

$$\begin{aligned} \int_a^b f(x) dx &= \int_{x_0}^{x_1} \left[\frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1) \right] dx \\ &\quad + \frac{1}{2} \int_{x_0}^{x_1} f''(\xi(x)) (x - x_0)(x - x_1) dx. \end{aligned} \tag{4.23}$$

$a < z < b$

$$\begin{aligned}
 \int_{x_0}^{x_1} f''(\xi(x))(x-x_0)(x-x_1) dx &= f''(\xi) \int_{x_0}^{x_1} (x-x_0)(x-x_1) dx \\
 &= f''(\xi) \left[\frac{x^3}{3} - \frac{(x_1+x_0)}{2}x^2 + x_0x_1x \right]_{x_0}^{x_1} \\
 &= -\frac{h^3}{6} f''(\xi).
 \end{aligned}$$

$x^2 - x x_1 - x x_0 + x_0 x_1$
 $\int 3 dx \rightarrow 3x$
 $h = x_1 - x_0$

Consequently, Eq. (4.23) implies that

$$\begin{aligned}
 \int_a^b f(x) dx &= \left[\frac{(x-x_1)^2}{2(x_0-x_1)} f(x_0) + \frac{(x-x_0)^2}{2(x_1-x_0)} f(x_1) \right]_{x_0}^{x_1} - \frac{h^3}{12} f''(\xi) \\
 &= \frac{(x_1-x_0)}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi).
 \end{aligned}$$

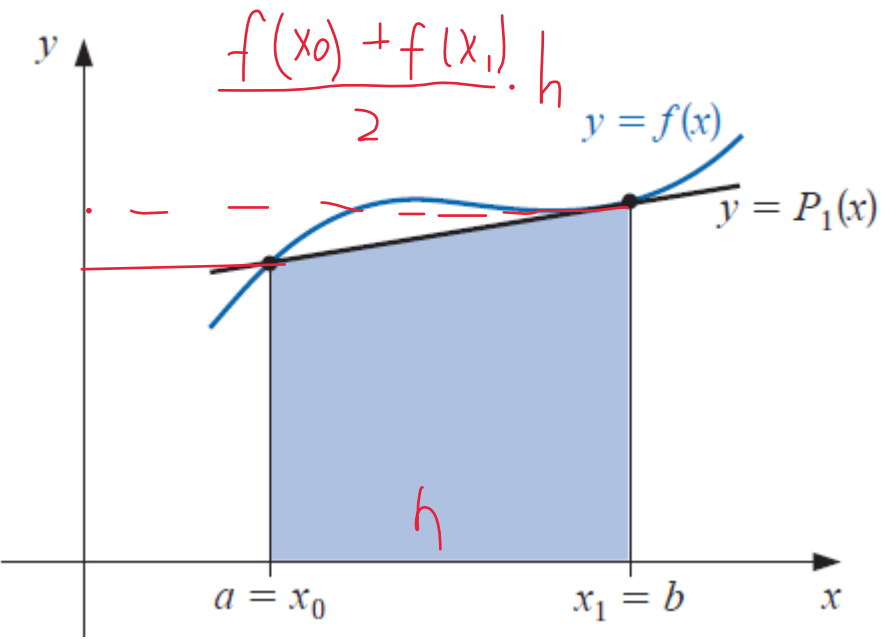
$x = x_1 : \frac{(x_1-x_0)^2}{2(x_1-x_0)}$
 $x = x_0 : \frac{(x_0-x_1)^2}{2(x_0-x_1)}$

Using the notation $h = x_1 - x_0$ gives the following rule:

Trapezoidal Rule:

$$\int_a^b f(x) dx = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi).$$

This is called the Trapezoidal rule because when f is a function with positive values, $\int_a^b f(x) dx$ is approximated by the area in a trapezoid.



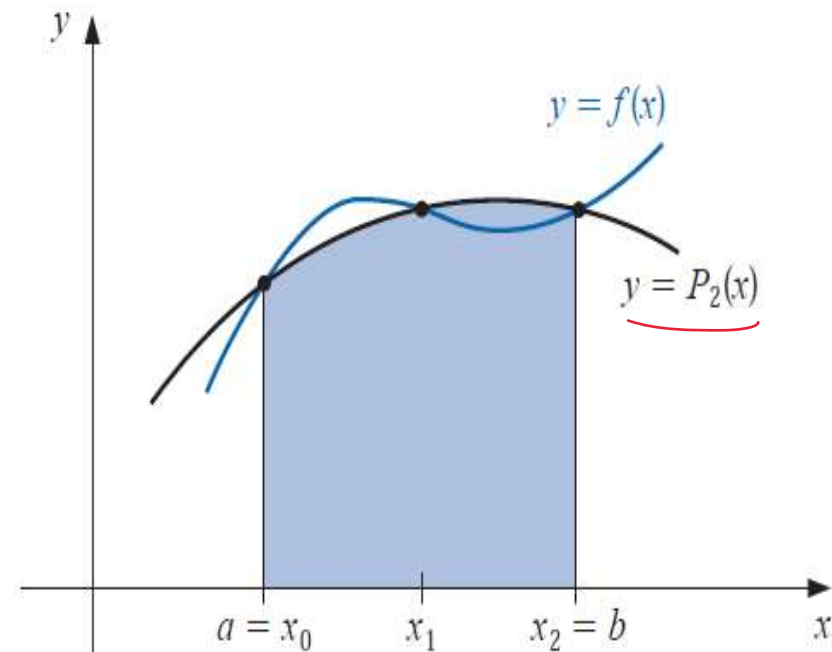
The error term for the Trapezoidal rule involves f'' , so the rule gives the exact result when applied to any function whose second derivative is identically zero, that is, any polynomial of degree one or less.

Simpson's Rule (Simpson Kuralı)

Simpson's rule results from integrating over $[a, b]$ the second Lagrange polynomial with equally-spaced nodes $x_0 = a$, $x_2 = b$, and $x_1 = a + h$, where $h = (b - a)/2$.

Therefore

$$\int_a^b f(x) dx = \int_{x_0}^{x_2} \left[\overset{L_0}{\frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0)} + \overset{L_1}{\frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1)} + \overset{L_2}{\frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2)} \right] dx + \int_{x_0}^{x_2} \frac{(x - x_0)(x - x_1)(x - x_2)}{6} f^{(3)}(\xi(x)) dx.$$



Error term: $\frac{1}{24} \int_{x_0}^{x_2} f^{(4)}(\xi(x))(x - x_1)^4 dx$.

Simpson's Rule:

$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90} f^{(4)}(\xi).$$

The error term in Simpson's rule involves the fourth derivative of f , so it gives exact results when applied to any polynomial of degree three or less.

Example 1 Compare the Trapezoidal rule and Simpson's rule approximations to $\int_0^2 f(x) dx$ when $f(x)$ is

(a) x^2

(b) x^4

(c) $(x+1)^{-1}$

$a=0$

(d) $\sqrt{1+x^2}$

(e) $\sin x$

(f) e^x

$b=2$

Solution On $[0, 2]$ the Trapezoidal and Simpson's rule have the forms

$$\frac{h}{2} [f(x_0) + f(x_1)]$$

Trapezoid: $\int_0^2 f(x) dx \approx \frac{h}{2} [f(0) + f(2)]$ and $h=2$

Simpson's: $\int_0^2 f(x) dx \approx \frac{h}{3} [f(0) + 4f(1) + f(2)]$. $h = \frac{b-a}{2} = \frac{2-0}{2} = 1$

$h = 2 - 0 = 2$
 $x_0 = 0$ $x_1 = 1$ $x_2 = 2$

$= \frac{4 \cdot 1 + 4}{3} = \frac{8}{3}$

$$\frac{x^3}{3} \Big|_0^2 = \frac{8}{3}$$

When $f(x) = x^2$ they give

Trapezoid: $\int_0^2 f(x) dx \approx 0^2 + 2^2 = \underline{4}$ and

Simpson's: $\int_0^2 f(x) dx \approx \frac{1}{3}[(0^2) + 4 \cdot 1^2 + 2^2] = \underline{\frac{8}{3}}.$

The approximation from Simpson's rule is exact because its truncation error involves $f^{(4)}$, which is identically 0 when $f(x) = x^2$.

The results to three places for the functions are summarized in Table 4.7. Notice that in each instance Simpson's Rule is significantly superior. ■

$f(x)$	(a) x^2	(b) x^4 ^{x^5 5}	(c) $(x+1)^{-1}$	(d) $\sqrt{1+x^2}$	(e) $\sin x$	(f) e^x
Exact value	2.667	6.400 ^{$= \frac{2}{5}$ 5}	1.099	2.958	1.416	6.389
Trapezoidal	4.000	16.000	1.333	3.326	0.909	8.389
Simpson's	2.667	6.667	1.111	2.964	1.425	6.421